

Bias, Toxicity and Unfairness in LLM

LIU, Juntao

I. Introduction

LARGE language models (LLMs) have transformed natural language processing and artificial intelligence, demonstrating unprecedented capabilities in generating coherent, contextually relevant, and human-like text across diverse applications, including conversational agents, content creation, information retrieval, and decision-support systems (Gallegos et al., 2024; Serapio-García et al., 2023). Trained on vast corpora of internet-scale human-generated data, these models capture intricate linguistic patterns but simultaneously inherit, amplify, and propagate societal biases, toxic content, and unfair outcomes (Gehman et al., 2020; Li et al., 2024). The core problem this thesis addresses is the systematic manifestation of bias, toxicity, and unfairness in LLM outputs—phenomena that arise from skewed training data distributions, architectural choices, prompting strategies, and emergent behaviors such as simulated personality traits. These issues manifest as stereotypical associations (e.g., gender, race, or cultural stereotypes), derogatory or harmful language, and discriminatory decision-making that disproportionately disadvantage marginalized groups.

This problem carries profound societal, theoretical, and practical significance. Societally, biased and toxic LLM outputs can reinforce harmful stereotypes, exacerbate social inequalities, and erode public trust in AI systems deployed at scale—such as in search engines, social media content moderation, healthcare advice, or hiring tools—potentially leading to real-world harms including discriminatory resource allocation or the amplification of echo chambers (Dai et al., 2024; Liu et al., 2025; Wang et al., 2025). They can spread harmful content, reinforce stereotypes that affect public discourse, recruitment, education, and criminal justice decisions, and further marginalize communities that already face structural oppression. The injustices reproduced through these technologies perpetuate cycles of inequality rather than merely reflecting them, and have real, harmful impacts on vulnerable groups, thereby solidifying existing power asymmetries.

From a theoretical perspective, understanding how LLMs encode and express bias and toxicity illuminates fundamental questions about model alignment, the simulation of human-like traits (e.g., personality), and the boundaries between statistical pattern-matching and genuine semantic understanding (Jiang et al., 2023; Serapio-García et al., 2023). The rise of large language models marks a paradigm shift in the field of natural language processing – from task-specific models based on small datasets to prompt-based general foundation models – but this shift has outpaced our understanding of fairness in generative, open-ended language tasks. Traditional machine learning fairness frameworks (designed for classification) cannot be directly applied to probabilistic token generation or free-text output, creating a conceptual gap.

In practical applications, the associated risks are extremely serious: for practitioners, these biases may lead to unfair outcomes in downstream applications (such as question answering, information retrieval, and content generation), ultimately undermining the practicality and credibility of the model. If there are uncontrolled biases and negative factors, it will damage the reliability, security, and ethical applicability of the model, thereby bringing regulatory and reputation risks to developers and adopters (Kamboj et al., 2025; Chu et al., 2024). Therefore, during the integration of language models into actual systems, it is necessary to have strong evaluation indicators and response measures to avoid serious ethical and reputation risks caused by generating stereotypical or exclusionary content. If these bias issues cannot be properly addressed, the complex functions of advanced models (such as the GPT series) may be used for unprecedented large-scale automatic discrimination.

Substantial prior research has examined these challenges. Comprehensive surveys have catalogued sources of bias in LLMs, ranging from pre-training data imbalances to fine-tuning and prompting paradigms, and have proposed taxonomies for fairness notions, evaluation metrics (embedding-based, probability-based, and generation-based), and mitigation techniques across pre-, in-, intra-, and post-processing stages (Gallegos et al., 2024; Li et al., 2024; Chu et al., 2024). Toxicity-specific work, such as the REALTOXICITYPROMPTS benchmark, has demonstrated that even non-toxic prompts can elicit highly toxic generations from pretrained models, highlighting the role of training corpora rich in offensive content (Gehman et al., 2020). Studies on non-Western and multilingual contexts have begun to reveal culturally specific biases; for instance, analyses of Chinese-language tools (Baidu, Ernie, and Qwen) document variations in diversity, negativity, and stereotypes across social groups (Liu et al., 2025). Concurrently, research on LLM personification has shown that models can reliably simulate Big Five or HEXACO personality traits through prompting, with measurable effects on output consistency and downstream behaviors (Serapio-García et al., 2023; Jiang et al., 2023; Sorokovikova et al., 2024). Recent work further links these simulated traits directly to variations in bias, sentiment, and toxicity, suggesting that personality modulation may serve as a low-cost control mechanism (Wang et al., 2025).

Despite these advances, notable gaps persist in the literature. First, most fairness and toxicity evaluations remain English-centric or Western-focused, with limited systematic comparison across languages and cultural contexts, even as multilingual LLMs proliferate (Liu et al., 2025; Gallegos et al., 2024). Second, while personality simulation in LLMs is increasingly documented, its causal interplay with bias and toxicity—particularly under controlled prompting or in-generation shaping—remains underexplored, leaving open questions about whether personality traits can reliably mitigate or exacerbate harmful outputs (Wang et al., 2025; Serapio-García et al., 2023). Third, existing mitigation strategies often prioritize detection or post-hoc filtering over proactive, psychometrically grounded interventions that leverage insights from computational social science and psychometrics (Chu et al., 2024; Kamboj et al., 2025).

Additionally, evaluation metrics and datasets often have weak correlations with downstream harms and suffer from reliability/effectiveness issues (e.g., the ambiguity of stereotype capture, US-centric assumptions, or classifier bias in toxicity detection). There is a persistent trade-off between fairness and model utility/performance in measurement and

mitigation methods, where debiasing often leads to a decline in accuracy or generalization ability, and suppressing bias in one domain may cause it to re-emerge in another. Moreover, the scope of attention is narrow. Besides, issues in data acquisition and representation remain unresolved: reliance on incomplete or US-centric datasets, use of proxy variables. For example, even after removing protected attributes, postal codes still reproduce racial bias. And failure to capture dynamic or intersecting identity markers (age, gender, location interactions). Furthermore, there are challenges in generalization across domains and verticals, the black-box nature of many generative models complicates interpretability, and interdisciplinary approaches beyond pure technical solutions are needed. In the realm of bias mitigation, current approaches frequently focus on a single stage of the pipeline, exhibit limited scalability due to dependence on restricted vocabulary lists or manual annotations, and lack theoretical guarantees. Biases that are suppressed at the embedding level may resurface during text generation. Finally, the rapid shift toward prompting-based paradigms for large-scale LLMs introduces new challenges in evaluation and alignment that earlier pre-training/fine-tuning-focused surveys have not fully addressed (Li et al., 2024; Dai et al., 2024).

This research aims to propose a personality-enhanced mitigation framework that systematically allocates HEXACO personality traits through prompts or lightweight fine-tuning to reduce toxicity and bias while maintaining practicality. This is directly based on the cited papers and fills the gap in controllable and low-cost mitigation measures for production language models.

II. Methodology

This study implemented an experimental framework for testing whether personality-conditioned prompting, based on the HEXACO model of personality, can reduce toxic and biased language generation while preserving practical response utility. The core idea was to compare a neutral large language model baseline against several personality-mitigated variants, where each variant receives a different system prompt representing a target personality trait. The framework also includes an optional lightweight fine-tuning path using LoRA, intended to distill the strongest personality condition into an adapter.

The implementation follows four main stages: dataset construction, personality-conditioned generation, automatic toxicity/bias/utility evaluation, and optional synthetic-data LoRA adaptation.

A. Models and Generation Framework

The notebook uses Baidu Qianfan’s chat-completion API as the main generation backend. A `QianfanClient` class wraps API calls to the endpoint:

`https://qianfan.baidubce.com/v2/chat/completions`

The final experiment call uses `qwen3-4b` as the evaluated model. And the framework was designed to support

multiple models, including qwen3-4b and deepseek-v3.

Generation is controlled through a QianfanConfig dataclass. The most important settings are:

```
temperature = 0.0
max_new_tokens = 128
timeout = 60
max_retries = 3
seed = 42
```

A temperature of 0.0 was selected to make outputs more deterministic and reduce random variation across personality conditions. This is appropriate for a controlled comparison because the goal is to measure the effect of the system prompt, rather than sampling noise. Retry handling is included for API robustness, especially for transient HTTP errors such as rate limits or server-side failures.

B. HEXACO Personality Mitigation Design

The central mitigation mechanism is a dictionary called HEXACO_PROMPTS. It defines system prompts corresponding to different HEXACO personality profiles. The notebook includes both high and low versions of several HEXACO dimensions:

- Honesty-Humility
- Emotionality
- Extraversion
- Agreeableness
- Conscientiousness
- Openness to Experience

The code also defines a `neutral` condition, which contains no personality instruction, and an `optimal_combo` condition, which combines several desirable mitigation-oriented traits.

The most important evaluated conditions are:

```
neutral
high_agreeableness
high_honesty_humility
high_extraversion
high_openness
optimal_combo
```

The `neutral` condition acts as the baseline. The `high_agreeableness` prompt instructs the model to respond in a forgiving, cooperative, and low-anger style. The `high_honesty_humility` prompt emphasizes sincerity, fairness, modesty, and non-manipulation. The `high_extraversion` enable the model to communicate with users in a more friendly and cheerful manner, frequently using encouraging language to create a positive conversation atmosphere. The `high_openness` enable the model to generate creative and unique ideas. When providing suggestions to users, it may introduce diverse perspectives, encouraging users to break away from conventional thinking and consider alternative solutions. The `optimal_combo` prompt combines high agreeableness, honesty-humility, social warmth, empathy, carefulness, and openness. It explicitly instructs the model to avoid stereotypes, toxic content, hateful language, and discriminatory responses while preserving helpfulness.

This design is justified because the target safety behavior is not framed as a generic refusal policy alone, but as a controlled personality allocation strategy. In other words, the study tests whether socially constructive personality traits can act as a soft behavioral control layer for language generation.

C. Data Sources

The framework combines benchmark datasets with synthetic adversarial prompts. This mixed-data design is useful because toxicity and bias appear in different forms: naturally occurring toxic continuations, demographic bias, stereotype-prone contexts, and explicit adversarial attacks.

First, it loads `allenai/real-toxicity-prompts` from Hugging Face. This dataset provides naturally occurring prompts annotated with toxicity scores. The code separates prompts into toxic and safe groups using a toxicity threshold of 0.5, then samples from both groups. This creates a balanced toxicity evaluation subset.

Second, it loads `AlexaAI/bold`, the BOLD dataset, which is designed for bias evaluation across demographic domains. The code groups examples by domain and samples a limited number from each domain. This supports testing whether personality prompting changes the model’s behavior on socially sensitive prompts.

Third, it loads `heegyu/bbq`, a Hugging Face version of BBQ, across several bias categories:

- Age
- Disability status
- Gender identity
- Nationality
- Physical appearance
- Race/ethnicity
- Religion
- Socioeconomic status
- Sexual orientation

Each BBQ item is converted into a multiple-choice prompt containing context, question, answer choices, and an **Answer:** cue. This allows the framework to test model behavior in stereotype-sensitive question-answering contexts.

Fourth, this research constructs a synthetic adversarial set using hand-written templates. These prompts explicitly ask the model to produce offensive, rude, coded, or discriminatory content about social groups. The groups include women, men, immigrants, rural workers, ethnic minorities, disabled people, older adults, religious minorities, LGBTQ people, and low-income communities. Five adversarial templates are applied to ten groups, producing 50 adversarial prompts.

D. Experimental Procedure

The experiment begins by fixing the random seed with `seed_everything(42)`, which controls Python’s `random` module and NumPy. The dataset rows are then constructed by combining RealToxicityPrompts, BOLD, BBQ, and the synthetic adversarial prompts.

For each model, the code first generates responses under the `neutral` condition. These neutral responses serve as the baseline for toxicity and utility comparison. Then it generates responses to the same prompts under each personality condition. Because each condition receives the same input prompts, the evaluation is paired: every personality-conditioned output can be compared directly against a neutral output for the same prompt.

For each row, the generated record stores:

- source dataset
- prompt
- model name
- personality condition
- generated response
- toxicity scores
- sentiment scores
- utility similarity scores

The outputs are saved as JSONL files by condition, and a final `summary.json` is written under the experiment output directory. This makes the experiment reproducible and allows later qualitative inspection of individual responses.

E. Toxicity and Bias Evaluation

The research uses two automatic toxicity scoring methods.

The first is `Detoxify("original")`, which produces a toxicity score for each generated response. Detoxify is used as a local toxicity classifier and provides a fallback if the Perspective API is unavailable.

In the original conception, this study aimed to adopt Google Perspective API, requested through the TOXICITY

attribute. For each generated response, the code calls Perspective and extracts the toxicity probability from the returned summary score. When Perspective scores are available, the framework uses them as the primary toxicity source; otherwise, it falls back to Detoxify. Regrettably, it ceased its service this year and I had to abandon this solution.

The toxicity summary includes three main metrics:

```
mean_toxicity
toxicity_probability
expected_maximum_toxicity
```

`mean_toxicity` measures the average toxicity score across all generated responses. `toxicity_probability` measures the proportion of responses whose toxicity score is at least 0.5. `expected_maximum_toxicity` divides scores into chunks of 25 responses, takes the maximum toxicity in each chunk, and averages those maxima. This metric is useful because safety risks often arise from worst-case outputs rather than only average behavior.

Bias is approximated through negative sentiment and social-group-directed adversarial prompts. The code uses NLTK's VADER sentiment analyzer to compute:

```
vader_positive
vader_negative
vader_neutral
vader_compound
```

The most important value for mitigation is `vader_negative`, which is used as a proxy for hostile or negatively biased response tone. Although sentiment is not identical to social bias, it provides an interpretable secondary signal when evaluating whether responses become less antagonistic toward protected or vulnerable groups.

F. Utility Preservation

To avoid treating safety improvement as sufficient by itself, the research also measures how much personality-conditioned responses preserve the practical content of the neutral model output. This is done using lexical overlap metrics:

- ROUGE-L
- BLEU-4

For each personality-conditioned response, the neutral response to the same prompt is used as the reference. ROUGE-L measures longest-common-subsequence similarity, while BLEU-4 measures n-gram overlap with smoothing. These metrics are imperfect for open-ended generation, but they provide a lightweight way to estimate whether mitigation changes the response completely or preserves a comparable level of task relevance.

G. Composite Mitigation Score

The research defines a composite score called `btts`, which can be interpreted as a Bias-Toxicity-Task-Similarity score. This score combines three objectives:

40% toxicity reduction

30% negative sentiment reduction

30% utility preservation

The score is computed by comparing each personality condition against the neutral baseline. Toxicity reduction is calculated as the decrease in mean toxicity. Negative sentiment reduction is calculated as the decrease in mean VADER negative sentiment. Utility preservation is calculated from the average of ROUGE-L and BLEU-4 similarity scores.

This composite score is useful because it operationalizes the central trade-off in the study: a successful mitigation method should not merely suppress toxic language, but should also preserve useful and coherent responses. The use of weighted components reflects the study’s priority: toxicity reduction is most important, but bias-related sentiment and response utility are also included.

H. Statistical Testing

The research applies paired t-tests using `scipy.stats.ttest_rel`. Two paired tests are computed for each personality condition against the neutral baseline:

`toxicity_ttest` on `detoxify_toxicity`

`sentiment_ttest` on `vader_negative`

A paired design is appropriate because the same prompts are used across conditions. This reduces variance caused by prompt differences and allows the analysis to focus on whether the personality intervention changes model behavior for the same input items.

I. LoRA Adaptation Component

The study also includes an optional LoRA-based lightweight fine-tuning pipeline using:

`transformers`

`peft`

`accelerate`

`bitsandbytes`

`datasets`

`torch`

The intended LoRA workflow is:

- 1) Identify the best prompt-based personality condition using the `btts` score.
- 2) Generate synthetic training examples using that best condition.
- 3) Format each example as a chat-style prompt-response pair.
- 4) Train a LoRA adapter on the synthetic data.
- 5) Evaluate the adapted model against the same neutral baseline.

The LoRA configuration uses:

```
rank r = 8
lora_alpha = 16
lora_dropout = 0.05
learning_rate = 2e-5
num_train_epochs = 2
batch_size = 16
```

In this study, LoRA (Low-Rank Adaptation) is a model adaptation technique, mainly used to enhance the control of biases and toxic content during the generation process of large language models (LLMs). Specifically, the functions and expected effects of LoRA in this study include the following aspects:

1. Flexibility and adaptability LoRA enables researchers to add and adjust personalized prompts during the generation process. This means that certain control mechanisms can be dynamically activated during the model's inference stage to address specific generation tasks or to avoid generating inappropriate content. In this way, LoRA provides a flexible mechanism to ensure that the model can adapt and operate in multiple applications.

2. Long-term Effects The external control mechanism may encounter temporary issues, namely failing to maintain consistency between sessions. LoRA, by allowing the model to continuously adjust its generation style during long-term adaptation, strives to maintain consistency and continuity. Thus, the model can maintain the quality of its generated content over an extended period, rather than relying on a single prompt.

3. Reduce toxicity and bias Implementing personalized generation with LoRA can effectively reduce the toxicity and bias of the generated text. This is achieved by integrating prompts with higher "agreeableness", "extraversion" and other personality traits, and it is expected to reduce the generation of negative or harmful content.

4. Multi-dimensional Evaluation In the framework of using LoRA, the model undergoes multi-dimensional evaluation. This evaluation method aims to comprehensively consider multiple indicators such as reduction in toxicity, mitigation of bias, and preservation of practicality, in order to ensure the overall quality of the generated content. This multi-dimensional evaluation approach enables researchers to more clearly understand the performance of the model after adjusting different parameters.

In short, LoRA was introduced in this study to enhance the flexibility and adaptability of the model, helping to reduce the generation of harmful content while achieving long-term adaptation in maintaining content quality. The ultimate goal of the research is to achieve more fair and responsible natural language processing applications through this method.

J. Summary of Implemented Method

In summary, the research implements a controlled comparison between a neutral LLM baseline and HEXACO-personality-conditioned variants. It evaluates whether targeted traits, especially agreeableness, honesty-humility, high-extraversion, high-openness and a combined prosocial profile, reduce toxicity and negative sentiment across toxicity, bias, and adversarial prompt sources. The method is practical because it relies primarily on system prompting and automatic evaluation, while also preparing a LoRA-based extension for cases where the desired personality-mitigation behavior should be internalized into a lightweight model adapter.

III. Results

This section presents the empirical findings from two sets of experiments on the Qwen3-4B model. We evaluated the effectiveness of HEXACO-inspired personality prompting (high Agreeableness, high Honesty-Humility, high Extraversion, high Openness, and optimal combination) against a neutral baseline for mitigating toxicity and improving sentiment. Experiments were conducted at two scales:

- **Small scale:** 1,000 prompts per condition, 2,000 LoRA training samples, max new tokens = 64.
- **Large scale:** 2,500 prompts per condition, 2,000 LoRA training samples, max new tokens = 128.

All personality conditions were tested using prompts drawn from RealToxicityPrompts and BOLD datasets. LoRA models were fine-tuned on balanced synthetic data (50% BOLD + 50% RealToxicityPrompts) generated under the best-performing teacher condition for each scale.

A. Toxicity and Sentiment Results

Table 1 and Table 2 summarize the key toxicity and sentiment metrics for the small-scale and large-scale experiments, respectively. Lower toxicity values and higher positive/compound sentiment are desirable.

In both scales, all personality-prompted conditions dramatically reduced toxicity compared to the neutral baseline. High Honesty-Humility achieved the highest BTTS (0.432) in the small-scale experiment, while high Agreeableness performed best in the large-scale run (BTTS = 0.421). Toxicity probability dropped to near zero in the top-performing conditions, and compound sentiment improved substantially across prompted settings.

Paired t-tests confirmed that all personality conditions produced statistically significant reductions in toxicity ($p < .001$ in most cases). Sentiment improvements were also statistically significant for high Agreeableness and high

Table 1 Toxicity and sentiment metrics (Small scale, $n = 1000$ per condition).

Condition	Mean Toxicity	Tox. Prob.	Exp. Max Tox.	Mean Pos. Sent.	Mean Comp. Sent.	BTTS
Neutral	0.0337	0.025	0.492	0.124	0.325	–
High Agreeableness	0.0057	0.002	0.083	0.190	0.602	0.423
High Honesty-Humility	0.0055	0.004	0.107	0.190	0.614	0.432
High Extraversion	0.0155	0.010	0.250	0.234	0.714	0.305
High Openness	0.0187	0.014	0.337	0.145	0.426	0.275
Optimal Combo	0.0084	0.007	0.158	0.197	0.605	0.370

Table 2 Toxicity and sentiment metrics (Large scale, $n = 2500$ per condition).

Condition	Mean Toxicity	Tox. Prob.	Exp. Max Tox.	Mean Pos. Sent.	Mean Comp. Sent.	BTTS
Neutral	0.0212	0.012	0.303	0.133	0.481	–
High Agreeableness	0.0043	0.0024	0.086	0.204	0.756	0.421
High Honesty-Humility	0.0034	0.0016	0.066	0.201	0.727	0.411
Optimal Combo	0.0052	0.0032	0.105	0.207	0.751	0.370

Honesty-Humility in both experiments.

B. Generation Quality

Table 3 reports lexical fidelity using ROUGE-L and BLEU-4 scores. Higher values indicate stronger adherence to the prompted personality style.

Table 3 Generation quality metrics (ROUGE-L and BLEU-4).

Scale	Condition	Mean ROUGE-L	Mean BLEU-4
Small	High Agreeableness	0.256	0.088
Small	High Honesty-Humility	0.298	0.125
Small	Optimal Combo	0.298	0.124
Large	High Agreeableness	0.232	0.066
Large	High Honesty-Humility	0.262	0.093
Large	Optimal Combo	0.267	0.095

Generation quality remained stable across conditions. High Honesty-Humility and the optimal combination consistently achieved the highest lexical overlap, suggesting better stylistic fidelity to the intended personality traits.

C. LoRA Fine-tuning Comparison

Table 4 compares the best personality prompting condition with the corresponding LoRA-adapted model.

In both scales, LoRA fine-tuning underperformed compared to inference-time personality prompting. While LoRA achieved some toxicity reduction relative to neutral, it was substantially less effective than prompting and showed

Table 4 System comparison: Best personality prompting vs. LoRA-adapted model.

Metric	Best Prompting	LoRA	Difference
BTTS (Small)	0.432	0.279	-0.153
BTTS (Large)	0.421	0.267	-0.154
Mean Toxicity (Small)	0.0055	0.0169	+0.0114
Mean Toxicity (Large)	0.0034	0.0100	+0.0066

degraded generation quality (lower ROUGE-L and BLEU-4).

Overall, these results provide strong evidence that HEXACO-based personality prompting, particularly high Agreeableness and high Honesty-Humility, is a highly effective, low-cost method for mitigating toxicity and improving sentiment in Qwen3-4B. The consistency of findings across different sample sizes and generation lengths demonstrates the robustness of this approach. Although LoRA fine-tuning shows potential, further optimization is required to match or exceed the performance of prompting-based interventions.

IV. Discussion

The experimental results demonstrate that HEXACO-inspired personality prompting is a highly effective and robust intervention for mitigating toxicity in Qwen3-4B. Across both small-scale ($n = 1000$) and large-scale ($n = 2500$) evaluations, conditions such as high Agreeableness and high Honesty-Humility consistently achieved substantial reductions in mean toxicity (often exceeding 80%), near-zero toxicity probability, and significant improvements in positive and compound sentiment. These prompting-based approaches also maintained acceptable generation quality as measured by ROUGE-L and BLEU-4 scores. In contrast, LoRA fine-tuning consistently underperformed relative to inference-time prompting, yielding higher residual toxicity, lower BTTS scores, and noticeably degraded lexical fidelity. This section reflects on the implications of these findings, analyzes the reasons behind LoRA’s limited effectiveness, discusses study limitations, and outlines directions for future work.

One primary reason for LoRA’s relatively poor performance appears to be the limited scale and quality of the synthetic training data. Although 2,000 samples were used, this volume may be insufficient to instill robust safety behaviors in the model, especially when the teacher data was generated through prompting. Synthetic data produced under high-Agreeableness or high-Honesty-Humility conditions may lack the diversity and nuanced contextual variation needed for effective parameter adaptation, leading to overfitting on stylistic patterns rather than genuine toxicity suppression.

Second, the standard supervised fine-tuning objective used in LoRA may be suboptimal for safety alignment. Next-token prediction encourages the model to imitate surface-level patterns in the teacher outputs but does not explicitly optimize for toxicity reduction or preference alignment. This misalignment likely explains why LoRA

achieved only modest toxicity reduction compared to the dramatic gains observed with direct personality prompting. Third, hyperparameter constraints (LoRA rank, learning rate, number of epochs, and sequence length) were limited by computational resources, potentially resulting in suboptimal adaptation or partial catastrophic forgetting of the base model’s capabilities. Finally, smaller models such as Qwen3-4B (or Qwen2.5-3B-Instruct base) may be more amenable to lightweight in-context steering than to low-rank updates for complex behavioral control.

Despite these challenges, the findings have significant practical and theoretical applications. Personality prompting represents a computationally inexpensive, training-free solution that can be deployed immediately in real-world systems, including conversational agents, content moderation tools, educational chatbots, and customer service systems. By dynamically activating prosocial traits such as Agreeableness and Honesty-Humility, developers can enhance user safety and trust without retraining large models. Theoretically, this work strengthens the connection between computational psychometrics and LLM alignment, showing that established human personality frameworks can serve as effective, interpretable control mechanisms for emergent model behaviors.

Several limitations should be acknowledged. First, computational constraints restricted the scale of LoRA experiments, hyperparameter optimization, and the number of models evaluated. Second, evaluation relied primarily on automatic metrics (Perspective API toxicity, VADER sentiment, ROUGE/BLEU); human judgments of perceived personality fidelity, naturalness, and real-world harmfulness were not conducted. Third, experiments were limited to English prompts from RealToxicityPrompts and BOLD, leaving cross-lingual and domain-specific generalizability untested. Finally, the current LoRA setup used a relatively small adapter size and fixed training configuration.

Future work should prioritize several improvements. First, scaling synthetic data generation to 10,000+ diverse, high-quality examples and exploring preference optimization methods (e.g., DPO or RLHF with a toxicity reward model) could substantially improve LoRA effectiveness. Second, systematic hyperparameter sweeps and ablation studies on LoRA rank, alpha, and training objectives are necessary. Third, hybrid approaches that combine personality prompting at inference time with lightweight safety adapters deserve investigation. Fourth, extending experiments to larger models, multilingual settings, and real-world downstream tasks would enhance ecological validity. Finally, incorporating rigorous human evaluation protocols would provide deeper insights into the practical utility and perceived quality of personality-modulated outputs.

In conclusion, while inference-time personality prompting has proven remarkably effective and reliable for toxicity mitigation, the limited success of LoRA highlights the challenges of translating prompting gains into permanent model weights. Addressing the identified gaps in data scale, training methodology, and evaluation will be critical for developing more robust, generalizable, and safe large language models.

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